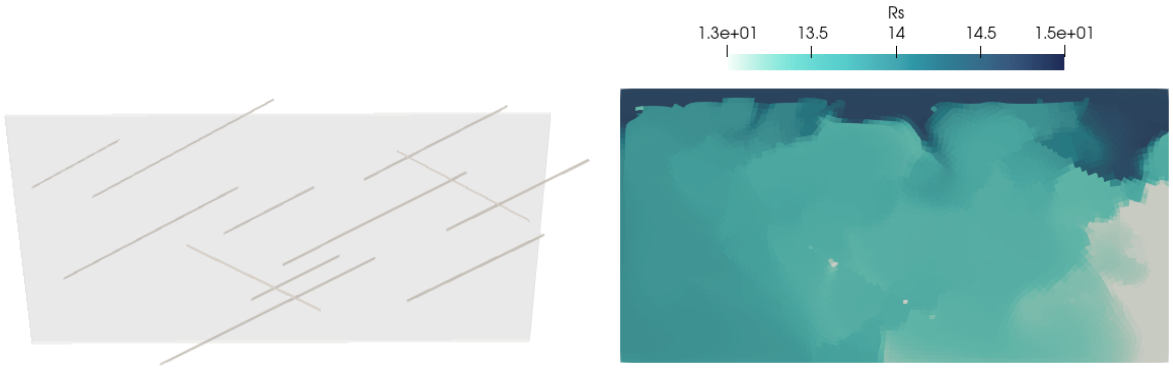
# DARSim Example:

## Fractures (pEDFM) 2D CCS




This example demonstrates how to use the pEDFM module in DARSim to simulate a fractured reservoir. Here, $CO_2$ is injected from the bottom left. The figure above shows the locations of the fractures (left) and the simulation results for dissolved $CO_2$ after approximately 125 years (right).

The properties and locations of the fractures are defined in `FractureInput.txt` along with various parameters to control the pEDFM methodology. For Cartesian and corner-point grids, the fracture generator must first be run with:

```
>> DARSimFracGen("path/to/FractureInput","FractureInput.txt")
```

This generates the fracture output, called `Fracture_Output.txt`. This file is then passed to `Sim.txt` with:

```
--Keyword for fractures
FRACTURED
      ON
      Fracture_Output.txt
```

For unstructured mesh types, the fracture generation and simulation are completed simultaneously when DARSim is called in MATLAB as usual. With unstructured meshes, the `Sim.txt` file simply defines the fracture input file with:

```
--Keyword for fractures
FRACTURED
      ON
      FractureInput.txt
```

Note that after the fractures are generated, the pEDFM information is saved in either Fracture_Output.txt (for Cartesian and corner-point grids) or fracture_process.mat for unstructured meshes. When making changes to FractureInput.txt, these files must be deleted before restarting the simulation.